Quantifying Pitcher Quality: Bayesian Regression on Run Value per 100 Pitches from Pitcher-Level Pitch Characteristics

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We develop a Bayesian Stuff+ index for MLB pitchers by modeling run value per 100 pitches as a function of pitch mechanics, plate location, and covariates. Aggregating 2023–2025 Statcast data to the pitcher x pitch-type x season level, we fit a joint Student-t regression with pitch-type-specific slopes and a pitcher-level random intercept via brms/Stan. The resulting Stuff+ scores track actual run-value outcomes across all eight pitch types. The framework applies to any pitch-tracking dataset, including minor league and international contexts.

1 Introduction

Run expectancy—the average number of runs a team is expected to score from a given base-out state through the end of the inning—is the basis of most modern pitch-value metrics (Tango, Lichtman, & Dolphin, 2006). The change in run expectancy on a given pitch (ΔRE) assigns each pitch an immediate value or cost. Averaging ΔRE across all pitches of a given type thrown by a pitcher over a full season and scaling to a per-100-pitch rate yields *run value per 100 pitches*: a season-level measure of the run-scoring value a pitcher generates or suppresses through each pitch in their arsenal. Because pitch-level ΔRE is discretely distributed rather than continuous, we aggregate to the *pitcher* $\times$ *pitch-type* level, where the central limit theorem motivates a location-scale likelihood; we use a Student-t to accommodate the heavy tails observed in the aggregated outcome.

ERA, WHIP, and other traditional evaluation metrics conflate a pitcher's true ability with defense quality, sequencing luck, and park effects. Statcast and its predecessors made it possible to decompose performance into physical pitch properties—velocity, movement, spin, release point—independent of these confounds. Stuff+ (Longenhagen & Petti, 2022) operationalized this idea by modeling pitch characteristics against expected outcomes, but how much each mechanical feature contributes to run value per 100 pitches, and whether those contributions differ across pitch types, remains an open question.

This study fits a joint Bayesian Student-t regression with pitcher random intercepts to the Jensen-Holm Statcast Era Pitches dataset (Holm, 2024; 2023–2025 MLB seasons) using `brms`/Stan (Burkner, 2017). The model produces a mechanics-plus-location Stuff+ index derived from posterior fitted values.

2 Methods

2.1 Data and Outcome

Pitch-tracking data from the 2023–2025 MLB regular seasons were obtained from the Jensen-Holm Statcast Era Pitches dataset (Holm, 2024). The unit of analysis is the **pitcher** $\times$ **pitch-type** $\times$ **season** combination. Pitches were retained only if they had complete data on all model predictors and belonged to one of eight pitch types: 4-Seam Fastball, Changeup, Curveball/KC, Cutter, Sinker, Slider, Split-Finger, and Sweeper. Pitcher–pitch-type–season cells with fewer than 50 pitches were dropped. The outcome is **run value per 100 pitches** (`rv100`): the average ΔRE across all pitches of a given type thrown by a pitcher in a given season, multiplied by 100. More negative values are better for the pitcher. Scaling to a per-100-pitch rate rather than modeling the cumulative sum places all pitcher–pitch-type cells on a common scale regardless of usage volume, so that low-usage and high-usage pitchers are evaluated on equal footing. Because the outcome directly averages individual-pitch ΔRE, base-out state and count context are already embedded in the per-pitch values and need not be modeled separately.

2.2 Predictors and Covariates

All continuous predictors are pitcher $\times$ pitch-type means, standardized within pitch type (mean 0, SD 1). **Pitch predictors:** release velocity, spin rate, horizontal break (`pfx_x`), vertical break (`pfx_z`), arm angle, release extension, a velocity $\times$ vertical break interaction (`speed_ivb_z`), and mean plate location (`plate_x`, `plate_z`). **Covariates:** left-handed pitcher indicator (`lhp`) and proportion of left-handed batters faced (`prop_lhb_z`). All predictors and covariates receive pitch-type-specific slopes.

2.3 Statistical Model

The model is a single Bayesian Student-t regression fit jointly across all eight pitch types. Each pitch type gets its own intercept and its own slopes for the mechanics + location predictors through `pitch_name` interactions. A pitcher-level random intercept $u_j \sim \mathcal{N}(0, \sigma_u^2)$ pools information across a pitcher's arsenal:

$$\texttt{rv100}_{j,t} = \alpha_t + \beta_t^\top \bar{\mathbf{x}}_{j,t} + \beta_{\text{lhp},t} \cdot \text{lhp}_j + \beta_{\text{lhb},t} \cdot \texttt{prop\_lhb\_z}_{j,t} + u_j + \varepsilon_{j,t}$$

Modeling the per-100-pitch rate eliminates the scale dependence on pitch count, removing the need for observation-level weights. The Student-t likelihood, with degrees of freedom ν estimated from the data, naturally downweights outlier pitcher-seasons that a Gaussian would force the model to accommodate. Priors are weakly informative: $\beta_k \sim \mathcal{N}(0, 4)$, $\alpha \sim \mathcal{N}(0, 4)$; the pitcher random-effect SD and residual SD use Exp(0.5) priors; the degrees of freedom use a Gamma(2, 0.1) prior. Sampling used 4 chains with 4,000 iterations (2,000 warm-up). The Stuff+ index is computed from posterior fitted values $\hat{\mu}_{j,t}$, standardized within pitch type: $\text{Stuff+}_{j,t} = 100 - 20 \times (\hat{\mu}_{j,t} - \bar{\hat{\mu}}_t)/\text{SD}(\hat{\mu}_t)$. A Stuff+ of 100 is league average; higher is better.

2.4 Assumptions and Assessment

Assumptions: (1) Conditional exchangeability within pitch type given all predictors. (2) A Student-t likelihood with estimated degrees of freedom is adequate at the pitcher level, accommodating heavier tails than a Gaussian while remaining justified by CLT-based aggregation over ≥ 50 pitches per cell. (3) The linear additive specification is a reasonable first-order approximation. (4) Predictor means are measured precisely enough at $n_{j,t} \geq 50$ to treat as fixed. (5) Residual variance of the per-100-pitch rate is approximately constant across cells; remaining heteroskedasticity from sample-size differences is mild after the ≥ 50 pitch filter. **Assessment:** Convergence is monitored via $\hat{R}$, bulk ESS, and divergent transitions. Posterior predictive checks evaluate the Student-t likelihood. Model fit is summarized by overall and per-pitch-type prediction R^2 (squared correlation between fitted and observed values), and feature importance by the signal-to-noise ratio $\text{SNR} = |\hat{\beta}|/\text{SD}(\beta)$; an SNR near 2 corresponds to the 95% credible interval just excluding zero. Assumption diagnostics (QQ-plots, residual plots) are in Appendix A.

3 Results

3.1 Convergence and Model Fit

Table 1: MCMC Convergence Diagnostics

Model	Max $\hat{R}$	Min Bulk ESS	N Divergent	$\hat{R}$ OK	ESS OK
Joint model (pitcher RE)	1.006	1246	0	TRUE	TRUE

The convergence diagnostics are reported in Table 1. The maximum $\hat{R}$ across all parameters is 1.006, with a minimum bulk ESS of 1246 and 0 divergent transition(s). The overall prediction R^2 (squared correlation between fitted and observed values) is 0.125; per-pitch-type R^2 values are in Appendix A. Posterior predictive checks (Figure 1) confirm the Student-t likelihood fits the observed distribution well. Assumption diagnostics are in Appendix A.

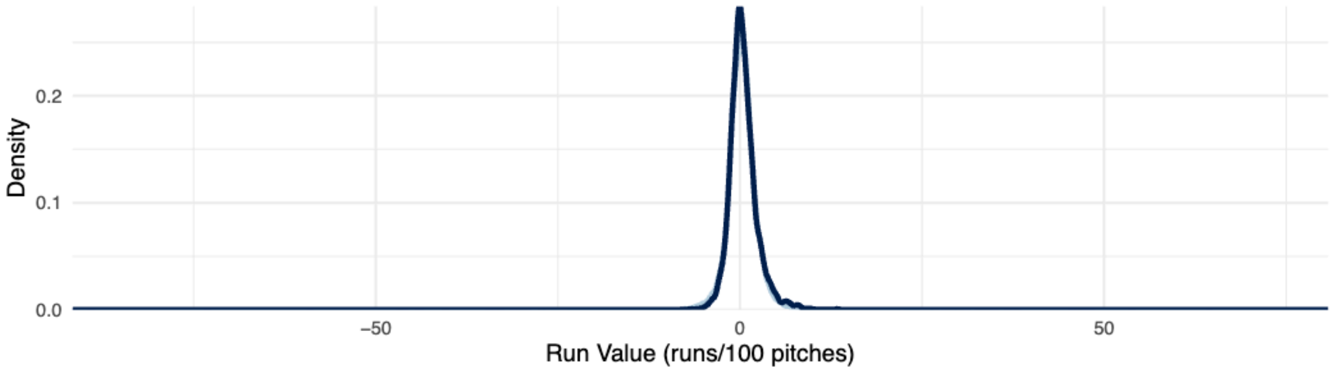

Figure 1: Posterior predictive check (50 draws, all pitch types combined).

3.2 Feature Importance by Pitch Type

To evaluate which pitch characteristics matter most—and whether the answer differs across pitch types—we compute the signal-to-noise ratio ($\text{SNR} = |\hat{\beta}|/\text{SD}(\beta)$) for each predictor from the joint model's pitch-type-specific interaction terms. An SNR above $\sim$2 means the 95% credible interval excludes zero.

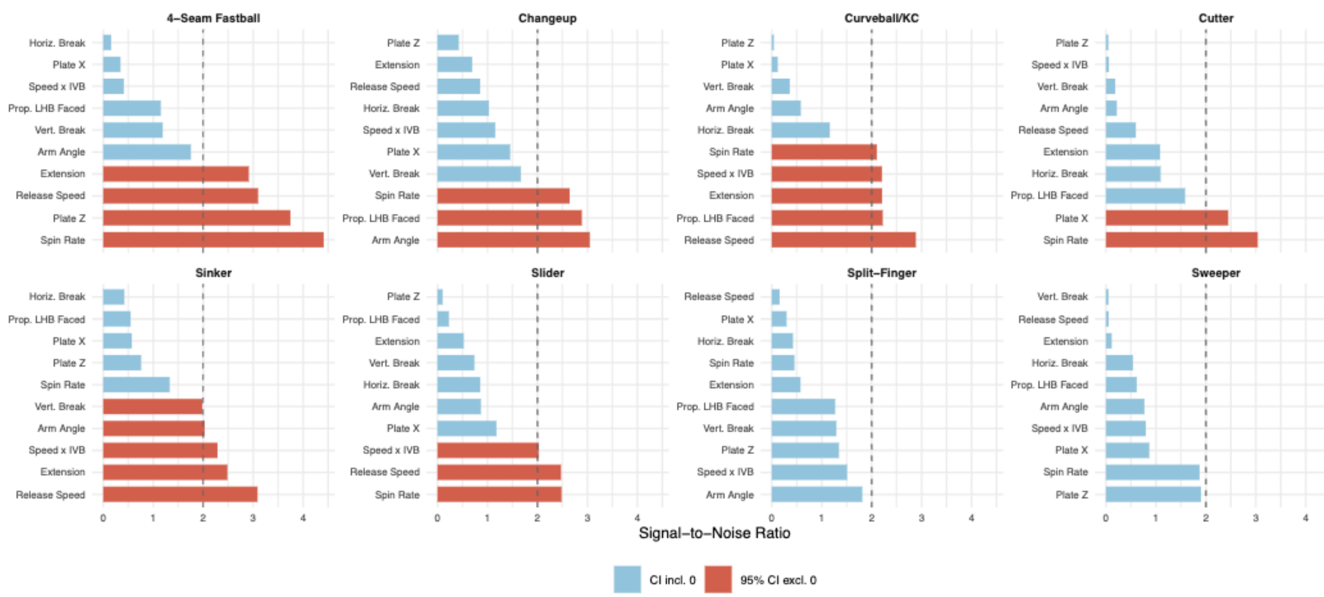

Figure 2: Signal-to-noise ratio for each predictor by pitch type. Red bars: 95% CI excludes zero. Dashed line: SNR = 2.

The SNR chart (Figure 2) shows which predictors matter most within each pitch type. Plate height (`plate_z`) emerges as a top predictor across most pitch types—particularly changeups, curveballs, cutters, and sliders—reflecting the importance of pitch location in the vertical plane. Horizontal break dominates for four-seam fastballs and sinkers, while the speed $\times$ IVB interaction and extension lead for sweepers and cutters. The proportion of left-handed batters faced is credible for sinkers and sliders, capturing platoon effects. Release speed is the strongest predictor for split-fingers, consistent with the pitch's reliance on velocity to create late downward action.

3.3 Variance Components and Stuff+ Derivation

The pitcher random intercept absorbs ICC = 0.079 of total variance (Figure 3), meaning that 7.9 percent of residual variation in run value per 100 pitches, after conditioning on pitch-type-specific mechanics and location, traces back

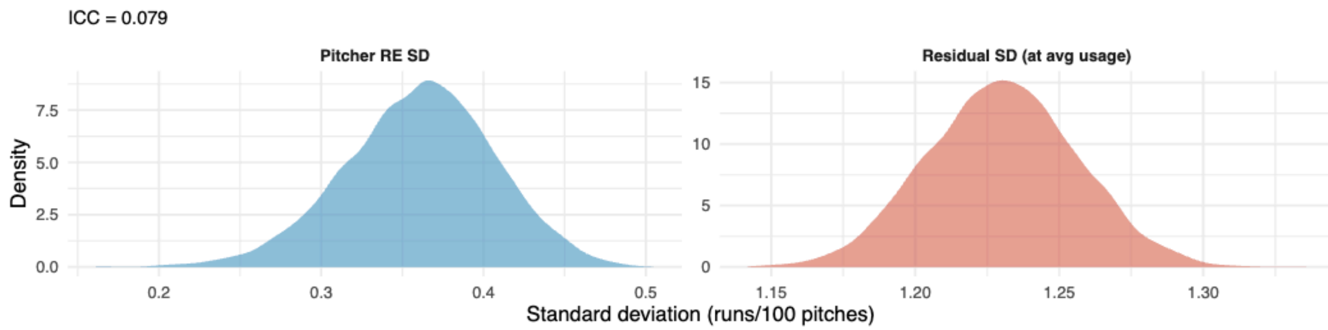

Figure 3: Posterior of pitcher random-effect SD and residual SD. ICC indicates the proportion of variance attributable to shared pitcher-level quality.

to a shared pitcher-level quality component that operates across a pitcher's entire arsenal.

3.4 Stuff+ Face Validity

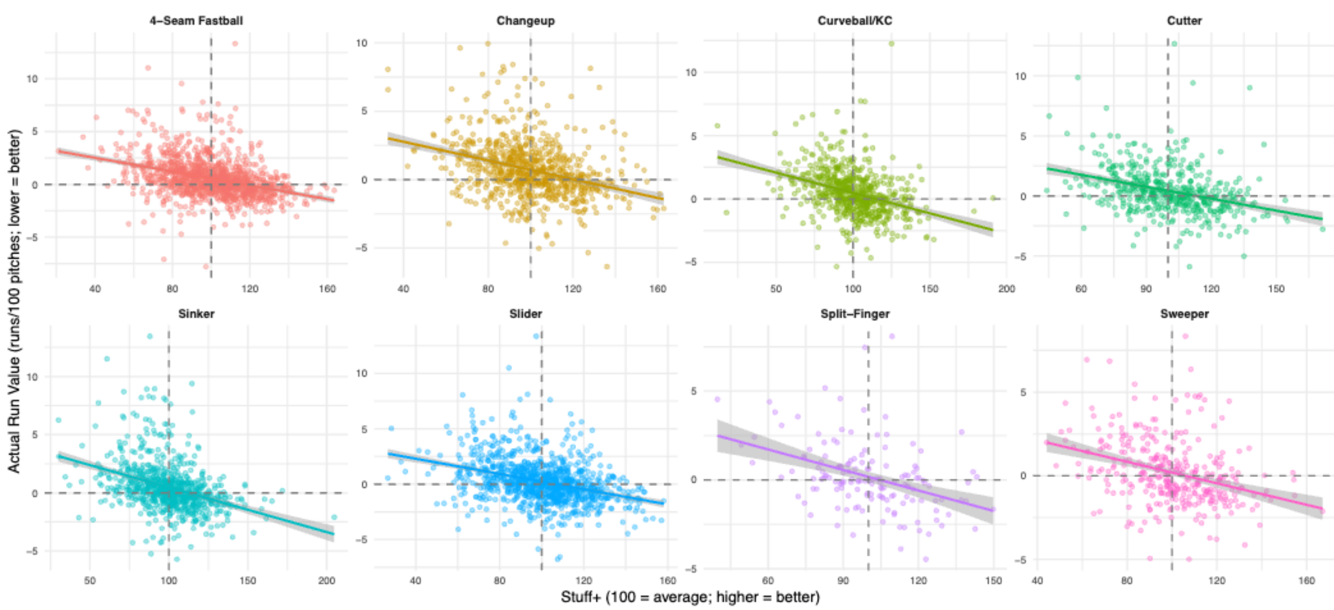

Figure 4: Stuff+ vs. actual run value per 100 pitches by pitch type. Higher Stuff+ corresponds to lower (better) run value across all eight pitch types.

Across all eight pitch types (Figure 4), higher Stuff+ tracks lower run value per 100 pitches. The negative linear trend is consistent across all pitch types, with tighter fits for the highest-volume types (four-seam fastballs, sliders, sinkers) and more dispersion for lower-volume types (split-fingers, sweepers), where fewer pitcher-season observations make estimates noisier. The partial pooling from the pitcher random intercept is doing the most work for these low-usage types, pulling extreme estimates toward the pitch-type mean. Top-ranked pitchers by Stuff+ are in Appendix A.

4 Discussion

The joint model produces a Stuff+ index that accounts for pitch mechanics, location, and pitcher-level quality simultaneously across three MLB seasons. The pitcher random intercept pools information across a pitcher's full repertoire, which is especially useful for low-usage pitch types where per-type sample sizes are small. The scatter plots confirm the expected negative relationship between Stuff+ and actual run value across all eight pitch types, and the top-ranked pitchers—Chapman (sinker), Clase (cutter), Yamamoto (split-finger)—pass a face-validity check against consensus evaluations of elite pitch quality.

The SNR analysis reveals that pitch location—particularly vertical height in the zone—is among the most credible predictors across pitch types, underscoring the role of command as a foundational pitching skill. Horizontal break is the dominant mechanics predictor for four-seam fastballs and sinkers, while extension and the speed × IVB interaction matter most for off-speed and breaking pitches. The proportion of left-handed batters faced is credible for sinkers and sliders, reflecting meaningful platoon effects that vary by pitch type. These findings support the practical conclusion that command development and pitch-type-specific mechanical adjustments offer the most direct paths to improving run-value outcomes.

The per-pitch-type prediction R^2 values (Appendix A) range from 0.114 to 0.164, indicating that pitch mechanics and location explain a modest but real share of variation in run value per 100 pitches. This is consistent with the inherent noisiness of pitch-level outcomes: even controlling for mechanics and location, per-pitch ΔRE is dominated by batter decisions, defensive alignment, game context, and sequencing effects that are not captured at the pitcher–pitch-type aggregation level. The primary purpose of the model is not to maximize R^2 but to produce a Stuff+ index that isolates the mechanics-driven component of pitcher quality from these confounds. The most impactful next step would be adding an explicit **opponent quality** covariate (e.g., opposing lineup wOBA), which would absorb the confound that pitchers facing weaker lineups accumulate better run values regardless of their own pitch quality.

Limitations. Pitch-type-specific slopes are fixed across pitchers and seasons; a random-slopes model would be more flexible but requires substantially more data per pitcher. The model treats each pitcher-season as an independent observation and does not capture within-pitcher development over time. Results are specific to the 2023–2025 seasons and the current Statcast classification system. Batter quality is only partially controlled through opponent handedness mix. Pitch sequencing and tunneling effects are not modeled—a release-point tunnel distance predictor or within-at-bat sequence features could capture deception effects in future work. Listwise deletion assumes MCAR, and the ≥ 50 pitch filter does not fully resolve selection effects for low-usage pitch types.

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5 AI Disclosure

Platform: Claude Sonnet (claude.ai), February–March 2026. **Use:** Debugging Quarto/R Markdown LaTeX rendering issues. **Transcript:** Available upon request per course requirements.

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Appendix A: Exploratory Data Analysis and Model Diagnostics

This appendix contains exploratory data analysis, model fit by pitch type, and assumption diagnostics referenced in the main text.

Table 2: Table A1: Pitcher-Season Descriptive Statistics — RHP

Pitch Type	N Pitcher-Seasons	Avg N Pitches	Mean RV/100	SD	Avg Velo (mph)	Avg Spin (rpm)	Avg IVB (in.)	Avg HB (in.)
4-Seam Fastball	1284	375	0.332	1.914	94.6	2307	15.5	-7.5
Changeup	637	209	0.515	2.388	86.5	1803	5.0	-14.4
Curveball/KC	546	227	0.361	1.875	80.4	2571	-10.0	8.9
Cutter	478	260	0.086	2.088	89.7	2407	7.8	2.3
Sinker	799	284	0.244	2.218	94.0	2194	8.0	-15.0
Slider	925	260	0.233	2.210	85.9	2436	1.6	5.1
Split-Finger	199	262	0.028	2.284	86.8	1338	3.3	-10.8
Sweeper	403	226	0.269	2.310	82.4	2600	1.4	14.0

Table 3: Table A2: Pitcher-Season Descriptive Statistics — LHP

Pitch Type	N Pitcher-Seasons	Avg N Pitches	Mean RV/100	SD	Avg Velo (mph)	Avg Spin (rpm)	Avg IVB (in.)	Avg HB (in.)
4-Seam Fastball	452	403	0.321	2.025	93.0	2245	15.3	7.8
Changeup	301	243	0.463	2.302	84.3	1735	5.6	14.1
Curveball/KC	196	246	0.372	2.107	78.6	2425	-8.9	-7.8
Cutter	157	224	0.380	2.004	87.8	2315	7.0	-1.9
Sinker	299	300	-0.037	2.199	92.4	2152	7.7	15.3
Slider	308	259	-0.042	2.157	84.5	2365	1.7	-4.6
Split-Finger	22	199	0.076	2.068	85.1	1206	5.3	8.6
Sweeper	136	217	-0.213	2.003	80.5	2496	0.7	-13.8

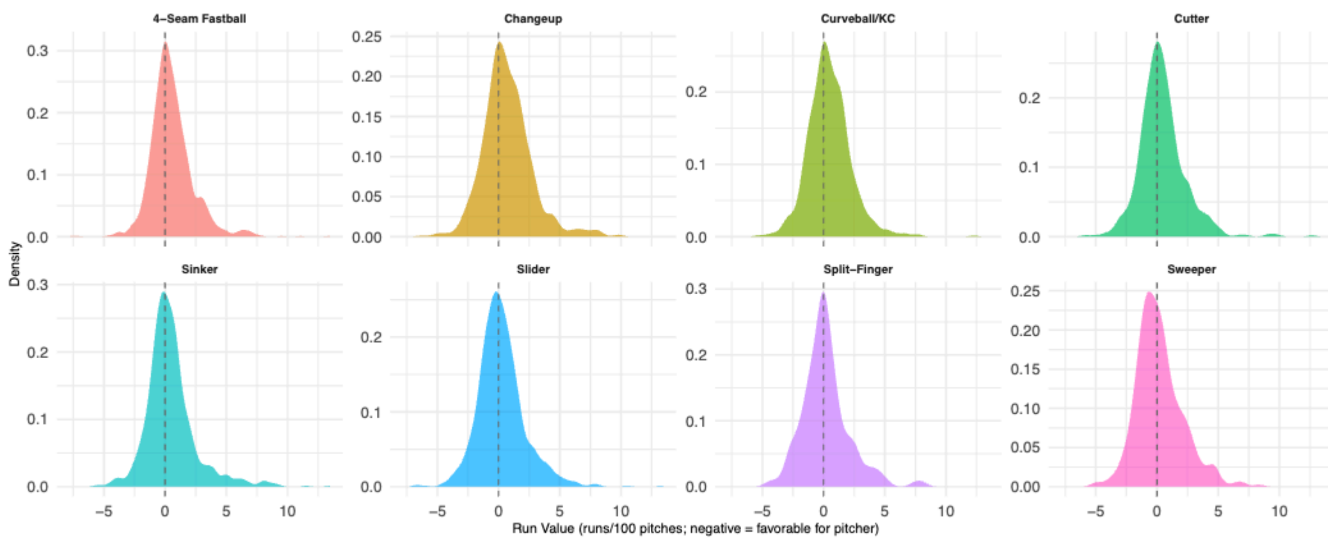

Figure 5: Distribution of pitcher-season run value per 100 pitches by pitch type.

Movement Profiles

Model Assumption Diagnostics

The following figures support the key model assumptions described in Section 2.4 of the main text.

Student-t likelihood adequacy (Assumption 2). Figure 5 above shows the outcome distribution is approximately symmetric and unimodal within each pitch type, with heavier tails than a Gaussian in some pitch types. The Student-t likelihood accommodates these tails while remaining consistent with CLT-based aggregation over ≥ 50 pitches. Figure 7 below provides formal QQ-plots.

Heteroskedasticity and pitch count (Assumption 5). Figure 8 shows residuals from the joint model versus standardized log pitch count. Modeling the per-100-pitch rate rather than the cumulative sum substantially reduces scale dependence on pitch count.

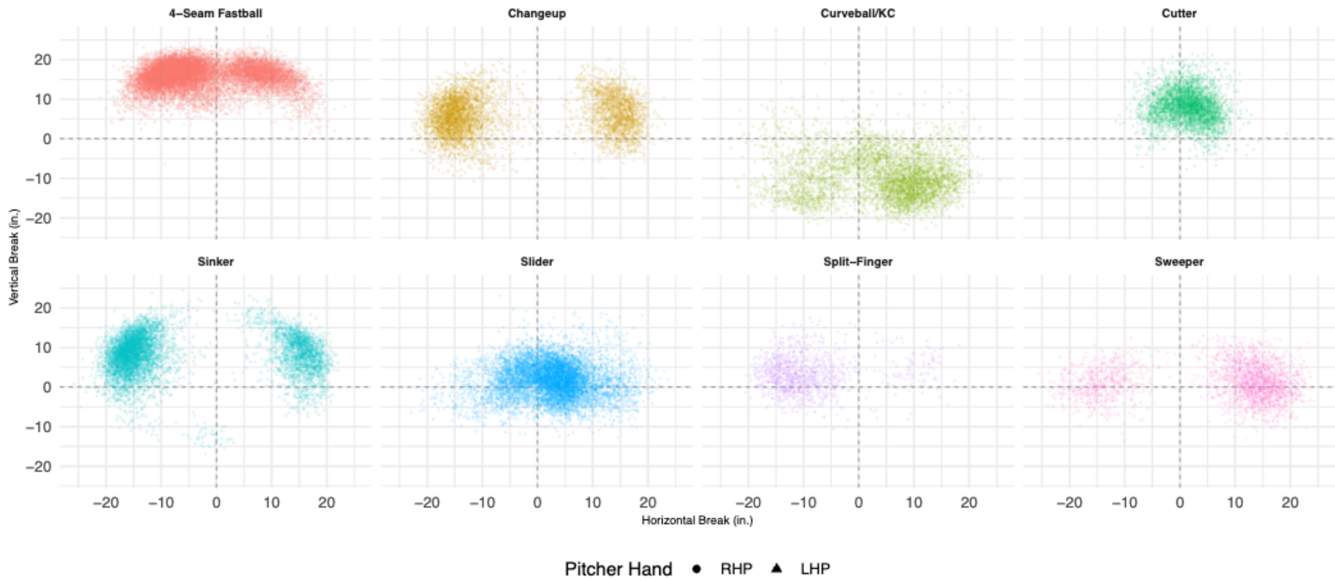

Figure 6: Movement profiles by pitch type (pfx_x vs. pfx_z). Shapes distinguish pitcher handedness. Clusters are well-separated, confirming clean Statcast classification.

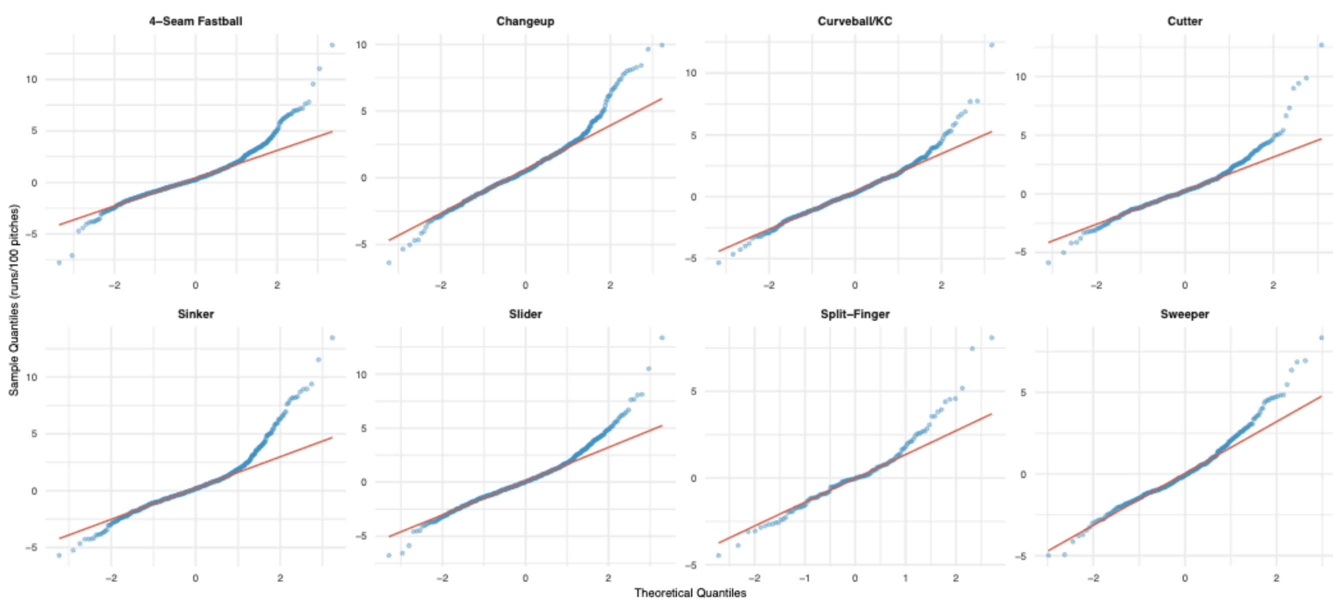

Figure 7: Normal QQ-plots of run value per 100 pitches by pitch type. Tail departures from the reference line motivate the Student-t likelihood.

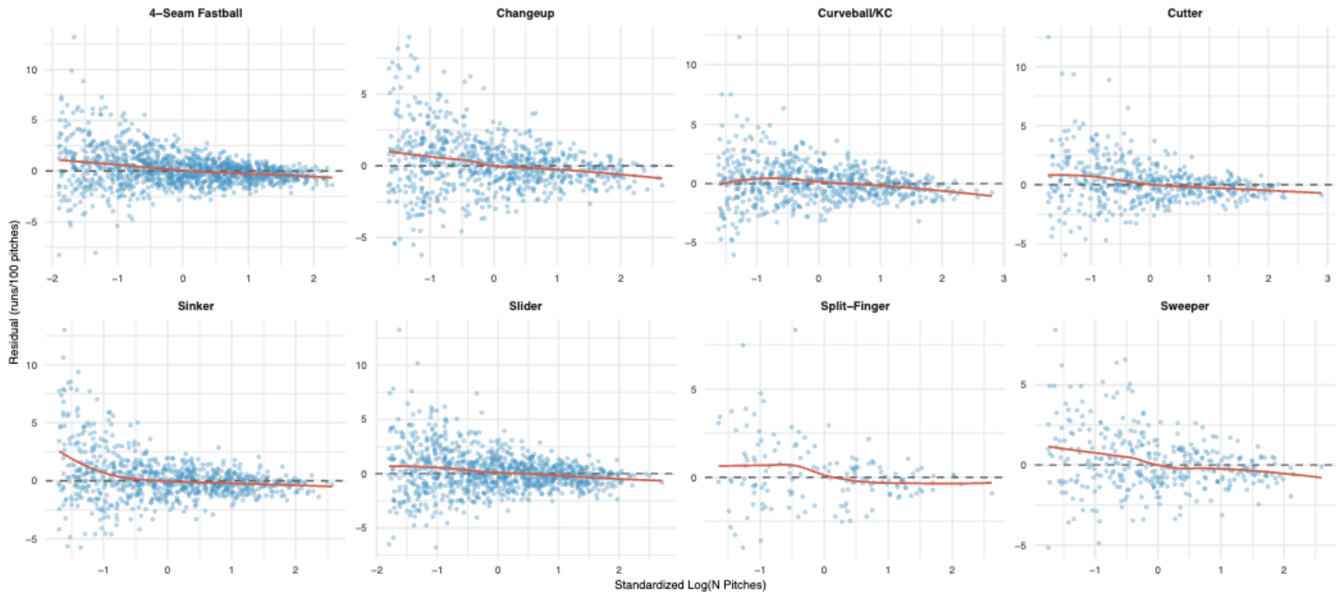

Figure 8: Residuals vs. standardized log pitch count by pitch type. Modeling the per-100-pitch rate reduces heteroskedasticity.

Linearity (Assumption 3). Figure 9 shows residuals versus fitted values from the joint model. No systematic curvature is apparent, supporting the linear additive specification.

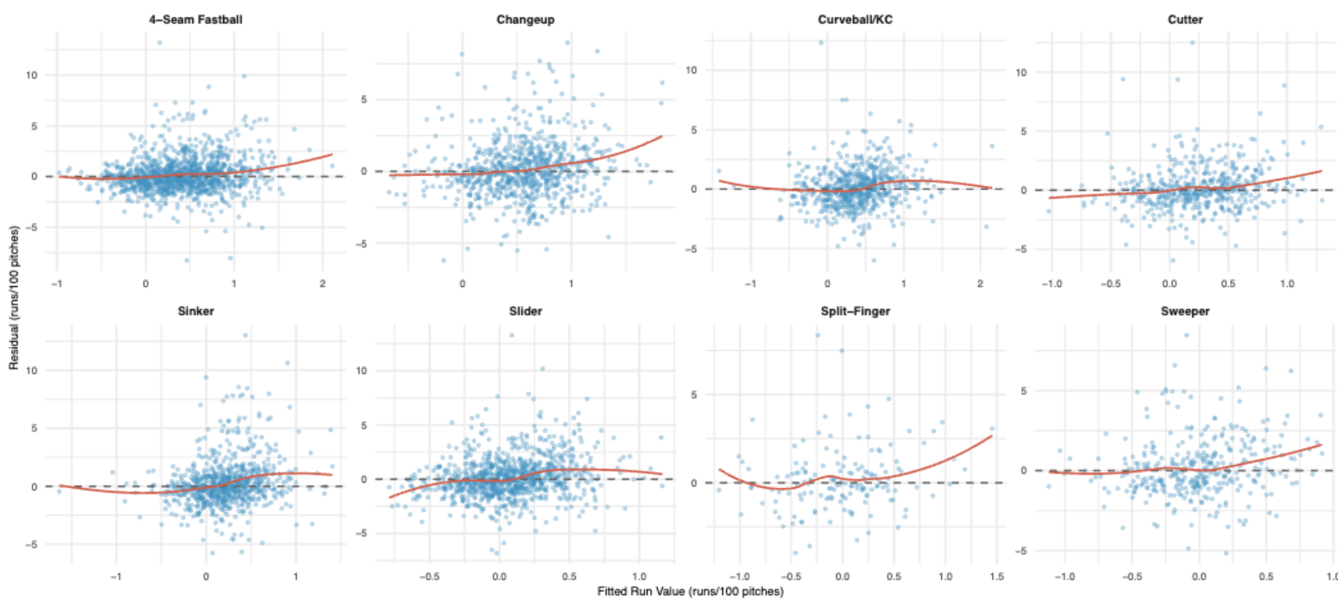

Figure 9: Residuals vs. fitted values by pitch type. The absence of systematic curvature supports the linear additive model specification.

Model Fit by Pitch Type

Stuff+ Rankings and Distribution

Pitcher Random Intercepts

Figure 10: Mean delta-RE by plate location and batter handedness. Pitches near the center of the strike zone have the highest (worst for pitcher) mean delta-RE.

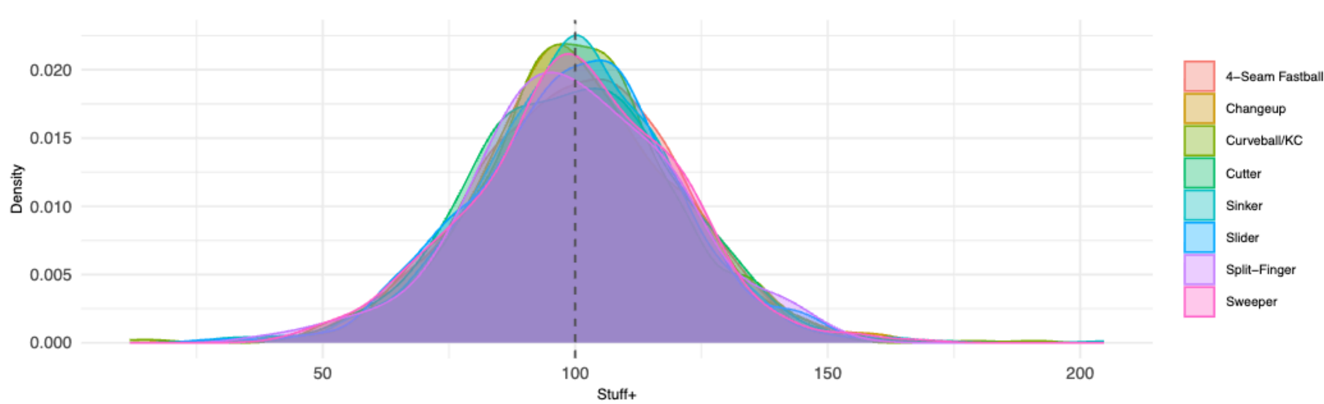

Figure 11: Distribution of Stuff+ across all eight pitch types.

Table 4: Prediction R^2 by Pitch Type (Joint Model)

Pitch Type	R^2
4-Seam Fastball	0.135
Changeup	0.114
Curveball/KC	0.128
Cutter	0.119
Sinker	0.138
Slider	0.128
Split-Finger	0.164
Sweeper	0.114

Table 5: Table A8: Top Stuff+ by pitch type (top 3, $\geq$100 pitches; 100 = avg, higher = better)

Pitch Type	Rank	Pitcher	Season	N Pitches	Actual RV/100	Stuff+
4-Seam Fastball	1	Miller, Mason	2025	478	-0.03	165.78
4-Seam Fastball	2	Misiorowski, Jacob	2025	613	-0.32	160.14
4-Seam Fastball	3	Wheeler, Zack	2025	849	-1.31	158.13
Changeup	1	Jax, Griffin	2024	175	-0.92	154.74
Changeup	2	Williams, Devin	2024	197	-0.76	154.64
Changeup	3	Richards, Trevor	2023	765	-1.02	152.57
Curveball/KC	1	Snell, Blake	2023	611	-3.49	158.43
Curveball/KC	2	Snell, Blake	2024	462	-2.08	154.50
Curveball/KC	3	Hentges, Sam	2024	112	-2.62	154.05
Cutter	1	Doval, Camilo	2024	377	0.39	174.06
Cutter	2	Clase, Emmanuel	2024	860	-2.80	171.72
Cutter	3	Clase, Emmanuel	2025	558	-1.29	170.02
Sinker	1	Chapman, Aroldis	2023	184	-3.35	210.27
Sinker	2	Chapman, Aroldis	2024	307	-0.21	207.02
Sinker	3	Chapman, Aroldis	2025	307	-2.30	206.99
Slider	1	Morejon, Adrian	2025	223	-1.70	160.43
Slider	2	Hall, DL	2025	101	-2.72	156.69
Slider	3	Scott, Tanner	2023	640	-2.45	154.99
Split-Finger	1	Yamamoto, Yoshinobu	2025	690	-2.10	142.34
Split-Finger	2	Yamamoto, Yoshinobu	2024	389	-0.87	141.84
Split-Finger	3	Bautista, Félix	2023	234	-3.29	138.66
Sweeper	1	Armstrong, Shawn	2025	244	-1.08	159.97
Sweeper	2	Coulombe, Danny	2023	143	-0.48	156.21
Sweeper	3	Coulombe, Danny	2025	119	0.14	156.19

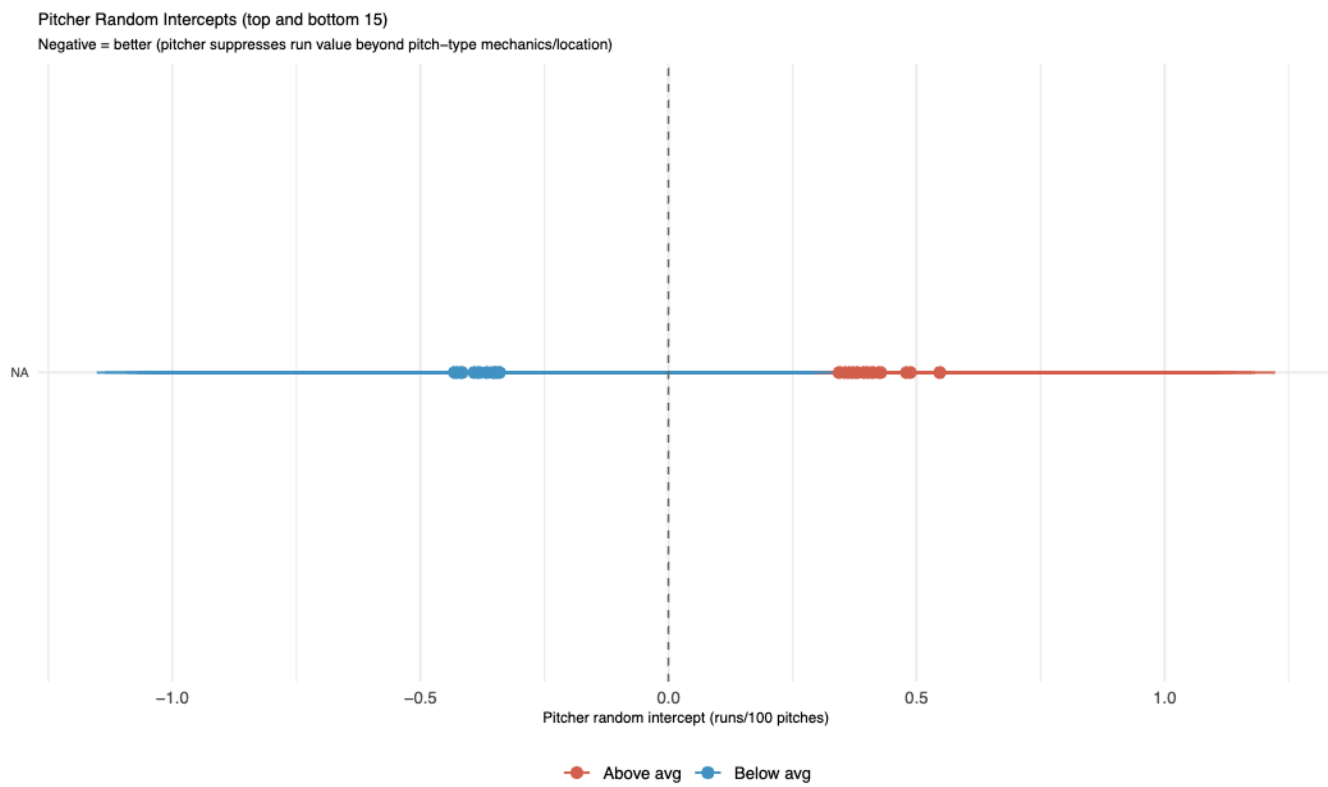

Figure 12: Top and bottom 15 pitcher random intercepts. Negative values indicate pitchers who suppress run value beyond what pitch-type mechanics and location predict.